Examining the Utility-based Resiliency of Utah's Highway Network

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Abstract

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1. Introduction

Systemic resiliency is an important consideration for transportation agencies, though specific definitions of "resiliency" might vary under different contexts. Some agencies and researchers see resiliency as facility-level design and engineering that hardens the system against failure (Bradley, 2007; Peeta et al., 2010); others as an ability for maintenance staff to rapidly restore service following catastrophe (Zhang and Wang, 2016); and others as an ability for a system to continue operating in degraded state (Berdica, 2002; Ip and Wang, 2011). Regardless of the definition used, assessing the resiliency of a transportation network — and addressing any potential shortfalls — requires a method to identify which links or facilities are most critical to the smooth operation of the network.

Identifying which links in a transportation network are most critical, however, is not trivial. State of the practice techniques typically rely on traffic volumes represented in terms of vehicles, trips, or freight value (AEM, 2017). But these methods ignore the fact that some networks have alternate routes readily available, have multiple well developed modes of transportation, or have redundant destination locations. When a network link does break, routes, mode choice, and destination choice often change too. These responses have been observed in real-world crisis events like the I-35 W bridge collapse in Minneapolis and the I-85 bridge fire and collapse in Atlanta (Zhu et al., 2010b,a; Xie and Levinson, 2011; Hamedi et al., 2018).

In this research, we present a model to identify critical highway links using destination choice accessibility analysis. This model is designed to capture the utility lost to individuals operating on a degraded network who may choose a longer route, a different travel mode, or a different destination. We then apply the model to a highway network representing the state of Utah. The model structure permits a more nuanced evaluation of link criticality than using traffic volume alone or in conjunction with an increased travel time measure.

The paper proceeds as follows. First, a literature review discusses previous attempts to analyze the resiliency of transportation networks. A methodology section presents the model developed for this research and describes the implementation process in Utah. A results section describes the model output in detail for one scenario and then compares and ranks the model outputs for an array of several dozen independent scenarios. The paper concludes with a discussion of limitations and associated avenues for future research.

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Table 1: Attempts to Evaluate Systemic Resiliency

Year	Author	Performance Metric
2004	Geurs and van Wee	Accessibility (isochrone, gravity, logsum)
2007	Abdel-Rahim et al.	Network Connectivity
2008	Taylor, M	Accessibility (logsum)
2010	Peeta et al.	Travel time and cost
2010	Geurs et al.	Accessibility (logsum)
2010	Levinson and Zhu	Travel time and cost
2010	Zhu et al.	Travel time and cost
2011	Agarwal et al.	Network connectivity
2011	Ip and Wang	Network connectivity
2011	Serulle et al.	Travel time and cost
2011	Ibrahim, S	Travel time and cost
2011	Xie and Levinson	Accessibility (isochrone)
2013	Omer et al.	Travel time and cost
2014	Osei-Asamoah and Lownes	Network connectivity
2015	Zhang et al.	Network connectivity
2015	Guze	Network connectivity
2015	Jaller et al.	Travel time and cost
2015	Xu et al.	Network connectivity
2016	Winkler, C.	Accessibility (gravity)
2017	Ganin et al.	Accessibility (gravity)
2019	Vodak et al.	Network connectivity
2019	Hackl and Adey	Network connectivity

2. Literature

In a groundbreaking theoretical article, Berdica (2002) attempted to identify, define and conceptualize network "vulnerability" — the complement of resiliency — by envisioning analyses conducted with several vulnerability performance measures including travel time, delay, congestion, serviceability, and accessibility. She then defined reliability as the level of reduced accessibility due to unfavorable operating conditions on the network. In particular, Berdica identified a need for further research toward developing a framework capable of investigating reliability of transportation networks.

In this section we examine several attempts by numerous researchers to do precisely this using various measures of network performance. It is helpful to categorize this review into three groups based on the overall technique applied in the study:

- *Network connectivity*: How does damage to a network diminish the connectivity between network nodes?
- *Travel time analysis*: How much do shortest path travel times between origins and destinations increase on a damaged network?
- *Accessibility analysis* How easily can the population using the damaged network complete their daily activities?

The following sections discuss relevant studies in each group; Table 1 consolidates these studies by year and labeled with applicable group.

2.1. Network Connectivity

The purpose of transportation networks is to connect locations to each other; presumably damage to a network would diminish the network's *connectivity*, or the number of paths between node pairs. It may even leave nodes or groups of nodes completely isolated. In studies based on connectivity, researchers typically apply methods and concepts from graph theory.

Abdel-Rahim et al. (2007) developed a multi-layered graph to examine the resiliency of the traffic signal control system in Boise, Idaho. The researchers determined which traffic signals would be isolated by a failure to a particular power substation, and consequentially the percent of travel paths that would experience diminished levels of service. The research highlights the degree to which interrelated infrastructure systems — power, telecommunications, and transportation — depend on each other, though the researchers did not attempt to look at the connective resiliency of the transportation network directly.

Agarwal et al. (2011) present a method to represent a transportation network as a hierarchical graph that can be analyzed more directly for vulnerabilities. The authors acknowledge, however, that a maximal failure consideration where a node is entirely isolated from the network is unlikely in a real-world network with multiple paths of connectivity.

Ip and Wang (2011) address this shortcoming through the concept of *friability*, or the reduction of capacity cased by removing a link or node, in order to determine criticality of individual links. The methodology relies on the ability to determine the weighted sum of the resilience of all nodes based on the weighted average of connectedness with other city nodes in the network. The authors determine that the recovery of transportability between two cities largely depends on redundant links between nodes. The authors also comment that most traffic managers are more concerned with the friability of single links rather than the friability of multiple links or an entire system.

Vodák et al. (2019) develop an approach to identify critical links in a network by searching for the shortest independent loops in the network. The algorithm progressively damages one or more links between iterations to determine if nodes become isolated, or cut off from the network. If a node becomes easily isolated or has a higher likelihood of becoming isolated, then there is a higher degree of vulnerability present in the network. This method can both identify critical links in individual networks, as well as provide a means to quantitatively compare networks.

All of these graph theoretical approaches tend to break down to some degree on large, real-world networks where the number of nodes and links numbers in the tens of thousands, and the degree of connectivity between any node pair is high. Highway system network failures — in most imaginable cases — degrade the shortest or least cost path but typically do not eliminate it entirely.

2.2. Changes in Travel Time

Graph theory based approaches to resiliency largely consider whether the network nodes become isolated after links are degraded or removed. Though isolation is an important problem, transportation networks often have multiple redundant paths between nodes; however, some of these paths may be considerably longer. It is therefore useful to determine how the path cost changes even if the nodes involved are still reachable on the network.

(Peeta et al., 2010) constructs a model to evaluate the most efficient allocation of highway maintenance resources prior to an earthquake with the potential to disrupt transport network connectivity. The authors create a simplified graph of Istanbul, Turkey with 30 attributed links and 25 nodes, with each link having a specified failure probability that could be mitigated with investment. Using this graph, the researchers evaluate what happened to travel times between the origin-destination pairs as links from this network were disabled using a Monte Carlo simulation. The authors showed that this problem is tractable with a locally optimal solution existing.

(Ibrahim et al., 2011) provides an alternative heuristic approach for determining vulnerability of infrastructure by estimating the cost of single link failure based on the increase in shortest path travel time due to increased congestion levels. Ibrahim proposes a hybrid heuristic approach that calculates the traditional user-equilibrium assignment for finding the first set of costs, and then fixes those costs for all following iterations to determine the effects of failure on overall travel time of the system. Ibrahim's novel heuristic

approach is important because of its ability to drastically reduce computation time for larger networks while providing accurate results that closely match results found from traditional modeling methods.

(Omer et al., 2013) proposes a methodology for assessing the resiliency of physical infrastructure during disruptions. To do this, the authors use a network model to build an origin-destination matrix that allows initial network loading and analysis. Omer's model uses several metrics, but the main metric used to determine resiliency is the difference in travel time between a disturbed and undisturbed network. Omer's framework is applied to an actual network between New York City and Boston for analysis. Changes in demand, travel time, mode choice and route choice are tracked for analysis. Omer's framework supports operability of transportation networks due to the way it analyzes networks experiencing suboptimal circumstances. The authors work identifies key parameters that should be measured to assess resiliency during disruptive events.

(Jaller et al., 2015) seeks to identify critical infrastructure based on increased travel time, or reduced capacity due to disaster. The proposed methodology utilizes user-equilibrium to determine proper initial network loading. Then, the shortest path between one origin and one destination can be identified. To implement damage to the network, a link is cut, and then the next shortest path is found. This process is followed for all links in the system in order to determine a sense of the criticality of each link to network resiliency. The analysis is carried out for each O-D pair, and the nodes with greatest change in travel time are determined to be the most critical. Jaller's methodology allows traffic managers to identify critical paths for mitigation purposes before the occurrence of disaster through careful analysis.

2.3. Changes in Accessibility

Accessibility refers to the ease with which individuals can reach the destinations that matter to them; this is an abstract idea but one that has been quantified in numerous ways. Dong et al. (2006) provide a helpful framework for understanding various quantitative definitions of accessibility that we will simplify here. The most elementary definition of accessibility is whether a destination is within an *isochrone*, or certain distance. This measure is often represented as a count, e.g., the number of jobs reachable from a particular location within thirty minutes travel time by a particular mode. Mathematically,

$$A_i = \sum_j X_j I_{ij}; I_{ij} = \begin{cases} 1 \text{ if } d_{ij} \leq D \\ 0 \text{ if } d_{ij} > D \end{cases} \tag{1}$$

where the accessibility A at point i is the sum of the all the jobs or other destinations X at other points j. I_{ij} is an indicator function equal to zero if the distance between the points $d_i j$ is less than some asserted threshold (e.g., thirty minutes of travel time). By relaxing the assumption of a binary isochrone and instead using the distance directly, we can derive the so-called gravity model,

$$A_i = \sum_j X_j f(d_{ij}) \tag{2}$$

where the function $f(d_{ij})$ is often a negative exponential with a calibrated impedance coefficient. An extension of the gravity model is to use the logsum term of a multinomial logit destination choice model,

$$A_i = ln \sum_j \beta_d(d_{ij}) + X_j \beta \tag{3}$$

Where the parameters β are estimated from choice surveys or calibrated to observed data. The logsum term has numerous benefits outlined by (Handy and Niemeier, 1997) and (**?**); namely, the measure is based in actual choice theory, and can include multiple destination types and travel times by multiple different modes. Accessibility measures of any kind are important in resiliency analysis because a damaged transport network will limit the ability of people to access the full variety of destinations they otherwise would.

? provide a review of accessibility measures such as those above up to 2004. Of the papers they reviewed, Vickerman (1974), Ben-Akiva and Lerman (1979), Geurs and Ritsema van Eck (2001) used isochrone type methods, Stewart (1947), Hansen (1959), Ingram (1971), and Vickerman (1971), and Anas (1983) used

gravity, and Neuburger (1971), Leonardi (1987), Williams and Senior (1978), Koenig(1980), Anas (1983), Ben-Akiva and Lerman (1985), Sweet (1997), Niemier (1997), Handy and Niemier (1997), Levine (1998), and Miller (1999) used or suggested logsums. They highlight the importance of using person-based measures such as these in evaluating network vulnerability and resiliency.

(Taylor, 2008) also examines accessibility as a way to analyze network vulnerabilities, mainly through logsum analysis. Taylor begins by explaining the concepts of vulnerability and accessibility, which further help to explain their interrelatedness and key distinctions. He then applies logsum analysis to provide a framework for solving problems of accessibility. Taylor's method for identifying vulnerabilities involves modeling travel demand, network topology, capacity, and road geometry in a manner that closely resembles a graph network. He does make one key distinction between links and nodes, however. A Node is vulnerable if link loss to that node diminishes its accessibility, while a Link is critical if its loss significantly diminishes accessibility within the network. One drawback to Taylor's methodology at this point, is that it is intended to be used after critical locations have been identified.

2.4. Summary

The lessons learned from the events in Minneapolis and Atlanta demonstrate that when transportation networks are damaged or degraded by link failure, multiple changes result. Traffic diverts to other facilities and other modes, and some people make fundamental changes to their daily activity patterns, choosing new destinations or eliminating trips entirely. Numerous other researchers have identified methodologies to capture the effects, or at least the costs, of these potential changes in modeled crisis events. We are able to learn from past and current methodologies to create a functional methodology on a state-wide level.

3. Methodology

This chapter presents the design and implementation of a model system designed to measure accessibility changes on a transportation network after the elimination or loss of highway links. The chapter begins with a discussion of the Utah Statewide Travel Model (USTM) and why it is not entirely suitable for this study. We then present a proposed model design and implementation within the CUBE travel demand modeling software. The chapter ends with a discussion of model calibration and validation.

3.1. Utah Statewide Travel Model

The Utah Statewide Travel Demand Model (USTM) is developed and maintained by the Travel Demand Modeling group at UDOT. Within Utah, there are five travel demand models developed for urban centers under the purview of a Metropolitan Planning Organization (MPO). USTM covers the rural and other areas lying outside of the MPO model regions, where UDOT is responsible for transportation planning. Consequently, USTM covers the entire state, and incorporates MPO models developed by the Wasatch Front Regional Council (WFRC), Mountainland Association of Governments (MAG), Cache Metropolitan Planning Organization (CMPO), and the Dixie Metropolitan Planning Organization (DMPO). Additionally, the Summit/Wasatch Travel Demand Model is incorporated into USTM (Utah Department of Transportation, 2021)

The local travel demand models and USTM use a gravity-based trip distribution model. The gravity model assumes that trips between origin-destination (OD) pairs are proportional to total productions P and attractions A. That is, all productions will be attracted to a location based on the size of a location (a locations attractiveness) and the friction factor between the OD pair. A mathematical representation of the gravity model is given by:

$$T_ij = P_i * (A_j F_{ij} / \sum A_j F_{ij}) \tag{4}$$

Where:

- T_{ij} represents trips made between an origin i and a destination j
- P_i represents productions at origin i

- A_j represents attractions at destination j
- F_{ij} is the friction factor between an OD pair

The friction factor (also known as the impedance, or resistance) between two zones can be represented in a number of ways, such as with a negative exponential function:

$$F_{ij} = exp(-d_{ij)} \tag{5}$$

Where d_{ij} is the distance or cost between zones i and j and , are calibrated parameters. In the gravity model, as the distance between an OD pair increases, users become less likely to make trips between that OD pair. Destinations are fixed, and trips calculated using the gravity model must have a proportional number of users assigned to each destination based on its size or level of attraction.

A primary weakness of gravity-based distribution models is their inability to consider multi-modal impedances or other attributes of a destination rather than their size (as represented by A_j in Equation (4)). The friction factor in Equation (5) asks an implicit question with its distance or cost variable d: which mode is used for the trip? In almost all cases, automobile distances are asserted as the only option, but if a destination happens to be close by rail and far by highway, the destination choice model will not incorporate this.

Alternatively to four-step models, logit-based destination choice models improve model accuracy when compared with gravity and four-step models, and are advantageous because of an increased ability to introduce additional variables and reflect other statistical assumptions (Travel Forecasting Resource, 2021). The ability to incorporate improved methods for determining trip distribution, mode choice, and destination choice all allow logit-based models to more accurately estimate trips between OD pairs on a road network. Consequently, logit-based travel demand model can estimate the choice-based effects of major changes to a road network given adverse natural or man-made events. More accuracy in a model's estimation process provides a more precise estimate than could otherwise be made using another type of model.

Destination choice models, on the other hand, explicitly consider multimodal accessibility. The ability to consider multimodal accessibility allows a user to choose the location of their destination based on a variety of factors including mode accessibility and other socioeconomic data, which are made available through the Utah Household Travel Survey (UHTS) conducted in 2017 (**?**). The socioeconomic data primarily comprises the size term of the destination choice. The DC size term is a measure of the appeal or attractiveness of a destination when compared to another destination. The DC size term is discussed in further detail later on in this paper.

A critical feature of logit-based choice models – described previously – is that they are more versatile than traditional modeling methods, with the ability to allow different types of data to be used for analysis which is highly important for a model that is constrained. Additionally, logit-based choice models are better able to measure the changes in accessibility of a destination due to network changes or disruption than other models. As such, logit-based models are typically used in new or more advanced travel models.

Such logit-based destination choice models are becoming common in four-step and more advanced travel models. However, logit-based destination choice models have not been implemented within the MPO models in Utah or within USTM except for some home-based work trips in the WFRC/MAG model. As a result, USTM and other travel models within Utah are not useful for the accessibility analysis proposed for this research, and a new model must be developed.

3.1.1. Model Framework

An initial model framework developed in 1 was created to show how a resiliency model would appear in order to create a logit-based model. The main inputs of the model are all extracted from USTM. The three main inputs to the framework are:

1. *Highway Network*: The USTM model network, which was obtained from UDOT and forms the basis of the model.
2. *Productions*: Zonal trip productions from the USTM model, classified as household productions.
3. *Attractions*: Zonal trip attractions from the USTM model, classified as socioeconomic data.

The model framework consists of the following steps (also visually represented in 1):

1. Skim Network - In this step, the model determines the shortest path by travel time on the USTM network between each origin and destination. The model also determines the corresponding distance.
2. Mode Choice Logsum

- The mode choice logsum is a function of the travel impedances estimated in the highway and peak transit skims and serves as an accessibility term in the destination choice model.
- The mode choice logsum uses utility functions to determine the probability and logsum associated with travel between each original and destination.
- The mode choice model uses constants and coefficients to calibrate and adjust the utility equations and determine the mode choice probabilities.

3. Destination Choice (DC) Logsum

- The destination choice logsum is a function of the travel impedances (represented by the mode choice logsum) and the attraction size term of each destination zone. This is the key evaluation metric of the model.
- The attraction size term is determined using socioeconomic data for each destination zone.

4. Destination Choice

- Allocates trip productions to destination zones based on information from the destination choice logsum.

5. Mode Choice

- Allocates trip productions attracted to each zone to specific modes based on the mode choice logsum.

6. Travel Time by Flow

- A new measure of travel time on the network after disruption used similarly to traditional network loading to measure effects of network disruption.

3.2. Highway and Transit Skims

The model requires an understanding of the distance between zones by multiple modes, and how these distances change when a link in the network is damaged or destroyed. To measure the automobile, transit, and non-motorized trip times, initial data was needed for each trip mode considered in the model.

The USTM highway network was extracted and applied to the resiliency model. The highway network is made up of both the urban and rural highway networks for the whole state of Utah. The highway network contains many link- and node-attribute data including street name, link distance, number of lanes, functional classification, transportation analysis zone ID (TAZ ID), county name, as well as speed limits and travel time data for five different times of day. Of particular interest from the available information was the AM_TIME which contains the travel time in minutes for the AM time period along a link and the $DISTANCE$, which contained the distance between nodes along a link. The AM_TIME is used to determine the travel time between an origin and destination. The $DISTANCE$ is used to measure the distance between the OD pair with the shortest AM travel time.

The highway skim module creates an output matrix of travel times and distances between origin-destination (OD) pairs and must be created before automobile and non-motorized (NMOT) trips can be incorporated into the model. We used the AM_TIME and $DISTANCE$ variables available in the highway network file to create a matrix of distances and shortest travel times between all OD pairs in the USTM network. The output matrix, or highway skim, forms the basis for further analysis by providing needed automobile and non-motorized (NMOT) information for the other modules in the model.

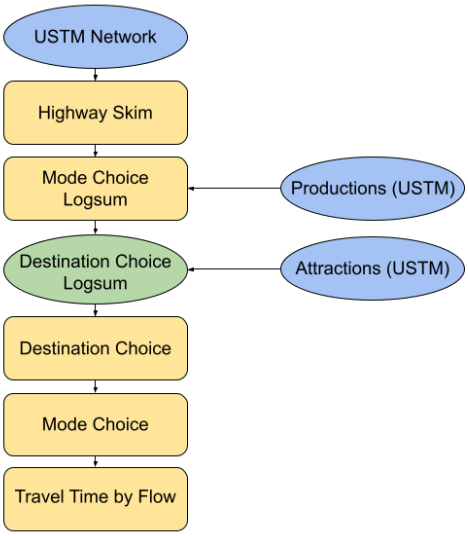

Figure 1: Model framework.

USTM incorporates traffic models maintained by other MPOs in Utah into the model structure. Only the WFRC / MAG model contains a substantive transit forecasting component. To account for transit trips in the resiliency model, an offline process adapted the WFRC transit skims into the USTM model zone structure. The end result of this process is that the TAZ matrices from each model were adjusted such that the WFRC / MAG model was the same size as the USTM model, and each TAZ number corresponded between the models as well.

Transit network resiliency is outside the scope of this project. Accordingly, the resiliency model assumes that transit services are fixed, meaning that changes to the network cannot influence transit availability on the network.

3.3. Trip Attractions and Socioeconomic Data

The resiliency model uses socioeconomic data extracted from USTM to estimate the productions at each zone. The socioeconomic data from the Utah Household Travel Survey (UHTS) conducted in 2015 contains TAZ related information such as county name, total households, household population, total employment, and a breakdown of employment by job category (**?**). This information is useful when determining the DC size term.

Trip attractions are calculated using the size term, which denotes the significance of a TAZ in attracting trips. The size term is built using various DC parameters and the socioeconomic data to determine the size or attractiveness of a zone. The size term equation was adapted from the Oregon Statewide Integration Model (SWIM), which is one of the most comprehensive destination choice models today.

3.4. Mode Choice Model

The mode choice (MC) module calculates the mode choice logsum (MCLS) between each OD pair in the network for each trip purpose. The trip purposes considered in the model are home-based work (HBW), home-based other (HBO) and non-home-based (NHB). The MC module includes the highway skim, the transit skim, and the MC coefficients and MC constants as inputs.

Table 2: Choice Model Coefficients

Variable	HBW	HBO	NHB
Destination Choice			
Households	0.0000	1.0187	0.2077
Office Employment	0.4568	0.4032	0.2816
Other Employment	1.6827	0.4032	0.2816
Retail Employment	0.6087	3.8138	5.1186
Distance	-0.0801	-0.1728	-0.1157
Distance^2	0.0026	0.0034	0.0035
Distance^3	0.0000	0.0000	0.0000
Mode Choice			
Shared	-1.1703	0.0164	-0.0336
Transit	-0.3903	-1.9811	-2.2714
Non-Motorized	-1.2258	-0.3834	-0.8655
Travel Time [minutes]	-0.0450	-0.0350	-0.0400
Travel Cost [dollars]	-0.0016	-0.0016	-0.0016
Walk Distance (less than 1 mile) [miles]	-0.0900	-0.0700	-0.0800
Walk Distance (1 mile or more) [miles]	-0.1350	-0.1050	-0.1200

The MC constants and coefficients used in the resiliency model were extracted from USTM where applicable or adapted from SWIM. The IVTT, COST, WALK1, and WALK2 coefficients for the HBW, HBO, and NHB purposes were extracted directly from the USTM mode choice model. The values for each of the coefficients are in Table **??**:

Utility equations, which were initially adapted from the RVTPO model, account for multiple variables which typically have different units. Each of the variables included in the utility equation must have the same unit, so the coefficients in Table **??** are made up of both generic and alternative-specific numbers. A generic coefficient is applied to a variable that changes by mode. In Table **??**, the only generic coefficient is the COST coefficient. The other MC coefficients in the resiliency model are alternative specific. These coefficients are the IVTT, WALK1, and WALK2 coefficients. The value that each of these coefficients are multiplied by changes for each of the available modes. The alternative specific coefficients included in the MC model allow the model to measure user bias.

Another aspect to utility equations are mode constants. We initially extracted the MC constants from USTM, however these values were adjusted during the calibration process. Constants are typically values that represent the effects of all factors on the choice, but are not limited to those values included in the utility equations (Koppelman and Bhat, 2006).

The utility equations from the Roanoke Valley Transportation Organization model (**?**) were extracted and then pared down to match the available data from USTM and the purposes included in the resiliency model. The constant values, which are represented using a "K" change during the calibration process while the coefficients do not change during the calibration process.

$$U_{Auto} = C_{IVTT} * TravelTime + C_{Cost} * Autocost * Distance \tag{6}$$

$$U_{Transit} = K_{TRN} + C_{IVTT} * TransitTime \tag{7}$$

$$U_{NMO}T = K_{NMOT} + 20 * C_{WALK1} * Dist_{<1Mile} + C_{WALK2} * Dist_{>1Mile} \tag{8}$$

From the equations above, several commonalities between the three MC utility equations are apparent. First, the transit and NMOT utility equations both have a constant, denoted by a K, included. The

auto equation does not have a constant because auto serves as the reference variable in the resiliency model. Second, both the auto and NMOT equations account for distance, and last, each of the three utility equations account for travel time, either in the form of an in-vehicle travel time coefficient, or another modified factor. The NMOT utility equation is not calculated using a specific coefficient for time. Instead, the NMOT distances are multiplied by an assumed walking speed of 20 minutes per mile. This is common practice in other choice models for NMOT trips.

Additionally, in the utility equations for NMOT trips, the maximum travel distance was limited to a trip of 50 miles or less. 50 miles was assumed as a maximum distance for a NMOT trip between an OD pair. We determined that the longest allowable distance a biker may choose to travel would be about 50 miles one-way. An attempt was made to limit the distance cap further; however, it was discovered that the NMOT distance becomes more sensitive at shorter distances. Limiting the NMOT trips will be explored further in Phase 2 of model development.

After the MC utilities were calculated, it was necessary to calculate the probability associated with each mode of travel for an OD pair. The equation used to accomplish this can be seen in the equation below:

$$Probability_i = \frac{exp_i}{\sum exp(U)} \tag{9}$$

Due to the way that the transit utility was calculated, the probability that a user would choose transit in an area with no available transit was often 100%. The cause of this was that the transit time matrix only shows a time value if there is transit available between an OD pair. The zero values that corresponded to areas with no transit availability caused the transit utility to be very small, or less negative in most cases compared to all the other modes considered. As a result, users had a greater probability of choosing transit as a mode even if it was not an available option between an OD pair. To troubleshoot this, we filtered the sum of the exponentiated utilities and reassigned values equal to zero to be equal to 1, so that the total sum of the exponentiated utilities for an OD pair would be 1. After implementing this change, the calculated probabilities added up correctly between modes, and areas with no transit service available returned a transit probability of zero.

3.5. Destination Choice Model

The destination choice (DC) module calculates the destination choice logsum (DCLS) between each OD pair in the network for each trip purpose. The trip purposes considered in the model are home-based work (HBW), home-based other (HBO) and non-home-based (NHB). Two other trip purposes are considered in USTM, though these purposes do not have significant effect on the logsum or are not flexible in nature (users cannot choose a different destination or origin). The DC module includes the highway skim, MCLS output from the previous module, the socioeconomic data extracted from USTM, and DC parameters as inputs.

The DC parameters were adapted where applicable from SWIM, the Oregon Statewide Integrated Model. SWIM has many more purposes and modes incorporated into its model structure than does the resiliency model. One limitation to using SWIM as a basis for the DC parameters, however, is that SWIM does not account for work trips. Oregon uses a separate travel demand model to estimate work trips, which was problematic because the DC parameters for work purposes could not be located. Thus, the work trip parameters were taken from the RVTPO model instead. The DC parameters can be seen in Table **??**

The DC utility equation was initially adapted from the RVTPO model, and then pared down to match the available data inputs from USTM and purposes of the resiliency model. The DC utility equation differs from the MC utility equations. The DC equation, seen in Equation (10), is made up of the MCLS calculated in the previous module, the size term, and several distance and cubic polynomial coefficients and their corresponding values.

$$U = C_{LSUM}*MCLS+log(sizeterm)+C_{Dist}*Dist+C_{Distsquared}*DistSquared+C_{DistCubed}*DistCubed+C_{LogDist}*log(Dist \tag{10}$$

The MCLS value is applied to the DC model utility equation as the impedance term, or a measure of a user's resistance to using the specified path or mode. This is like the friction factor in the gravity model

discussed in the beginning of methodology. Feeding the MCLS value into the DC module is what allows users to choose a destination rather than having fixed destination choices. The size term, which will be discussed in greater detail later in this chapter helps to determine the attractiveness of a destination zone compared to another destination. The cubic polynomial terms serve as a method to calibrate the DC module outputs and will also be discussed in greater detail later in this chapter. These are the main terms that make up the DC utility equation.

The socioeconomic data is used in the DC model to compute the size term, or the attractiveness of a destination choice in the model. The size term is made up of statistical data about a zone. This data was published in 2015 as part of the UHTS and made available via UDOT. The size term equation can be seen below:

$$SizeTerm = C_{HH}*TotHH+C_{OFFI}*OFFI+C_{RETEMP}*RETEMP+C_{OTHEMP}*(ALLEMP-OFFI-RETEMP) \tag{11}$$

Other employment was not included in the socioeconomic data, which was problematic. A way to determine a value for other employment was found by subtracting the other employment purposes considered from the ALLEMP value. Doing this left only the employment data that had not already been accounted for in each TAZ behind to be multiplied by the proper coefficient as seen in Table **??**.

The DC logsums are calculated by summing each row in the DC utility matrix, and then exponentiating that value. This process used to accomplish this can be seen below:

$$DCLS = exp(ROWSUM(U_{DC})) \tag{12}$$

The purpose of taking the log of the entire row is to measure the logsum between a zone and all the other zones at the same time. By doing this, we can determine the overall change in logsum between scenarios by TAZ, which is the measurement of interest. The logsum tells us the value of a zone based on a user's ability to choose a mode and destination.

The destination choice parameters and mode choice coefficients were calibrated according to standard procedure such that the trip length frequency distribution (TLFD) between USTM and the resiliency models were similar.

3.6. MAYBE INCLUDE TABLE WITH BEGINNING AND ENDING CALIBRATION VALUES?

3.7. Summary

We now have a logit-based model that is sensitive to user mode and destination choice. The model also accounts for modes that are not as flexible in the case of link closure as well. Using the model, we can input scenarios with broken links to evaluate the effects that link loss may have on user mode and destination choice, as well as estimate the overall disbenefit experienced by road users per day.

4. Results

In this chapter, we summarize the methodology used to identify vulnerable links on Utah's highway network. We also apply the model to scenarios where critical highway links are removed from the model network. This includes first a detailed analysis of a single scenario, where I-80 between Salt Lake and Tooele Counties is severed. We compare the model output to an alternative method that measures only the change in travel time and does not allow for mode or destination choice. The model was then applied to 41 individual link closure scenarios throughout the state.

4.1. Vulnerable Link Identification

Two methodologies were developed to identify vulnerable network links in Utah. The first method resembles the process used by AEM to determine threat categories, and threat proximity thresholds to highway links. This methodology was ultimately not used in the resiliency model. The second method uses an online Risk Priority Analysis map created by UDOT combined with familiar knowledge of Utah's road system. Using this second method, links were identified due to their location in relation populations or geography, are suspected choke points, or because they were of interest.

Table 3: Tooele Table

...1	Total Overall Costs	...3	Tooele - SLC Cost	...5	Base Trips
Purpose	Logsum	Travel Time	Logsum	Travel Time	Tooele - SLC
HBW	106106.36	244275.72	104645.23	233373.56	12143
HBO	27301.33	108412.94	23961.279999999999	98163.53	3578
NHB	40587.839999999997	84712.36	38817.089999999997	75361.98	5025
REC	398.72	398.72	40.729999999999997	40.729999999999997	3
XXP	55870.17	55870.17	$ -	$ -	0
Freight	10883835.01	10883835.01	111772.5	111772.5	515811
Total Comparable	173995.51999999999	437401.02	167423.6	406899.06	536560

Table 4: Tooele Table Again

Logsum / Base	...2	Logsum / Tooele - SLC Logsum	Travel Time / Tooele - SLC Travel Time
All Trips	Tooele - SLC		
0.43	0.45	0.99	0.96
0.25	0.24	0.88	0.91
0.48	0.52	0.96	0.89

4.2. Single Scenario Analysis

This section outlines an in-depth analysis that was conducted to ensure the resiliency model was accurately capturing trips with OD pairs in the targeted area around a closed link. This analysis was done on a link between Tooele and Salt Lake City, Utah.

Scenario 50, located along I-80 between Tooele and Salt Lake Counties, was examined to ensure the resiliency model was capturing trips in the vicinity of a broken link. Analyzing the model outputs at a localized level was necessary to ensure that the model was appropriately estimating trips and capturing them in the areas that were expected. In scenario 50, if many effected trips had not been captured in Tooele and Salt Lake counties, that would have signaled that a problem was present with the model. This analysis shows that a broken link mainly effects the area surrounding that link, and the key takeaway here is that the model functions as intended. Another method to estimate trip costs is the travel time method, which serves to capture trips that have more fixed OD pairs, such as freight trips. This analysis also looks at this method in addition to the logsum estimates.

Table 3 compares the overall costs between the logsum and travel time methods, the specific cost for trips originating in Tooele and ending in Salt Lake and includes a trip comparison as well. From Table 3, we can see that the logsum method captures about $173,996 in experienced expense due to the closure of the link. Specifically, between Tooele and SLC, the logsum based resiliency model captures $167,424 of expense, which is approximately 99% of the total expense as seen in Table 4. This shows that the resiliency model is effectively capturing trips in the correct areas based on the percent comparison capture rate in Table 4. When we look at the travel time method of analysis, we can see that the costs at both the local and statewide levels are much greater with $437,401 and $406,899 respectively estimated as the costs due to just the increase in travel time, not using the logit-based model.

The logsum and travel time methods for determining the overall cost associated with can be broken down into overall costs, and comparable costs. The comparable costs are made up of those purposes which are included in the resiliency model. HBW, HBO, and NHB trips are included in both models, thus the results from both the logsum model and the travel time method can be compared to each other.

The travel time method measures the difference in travel time between the base scenario and any other

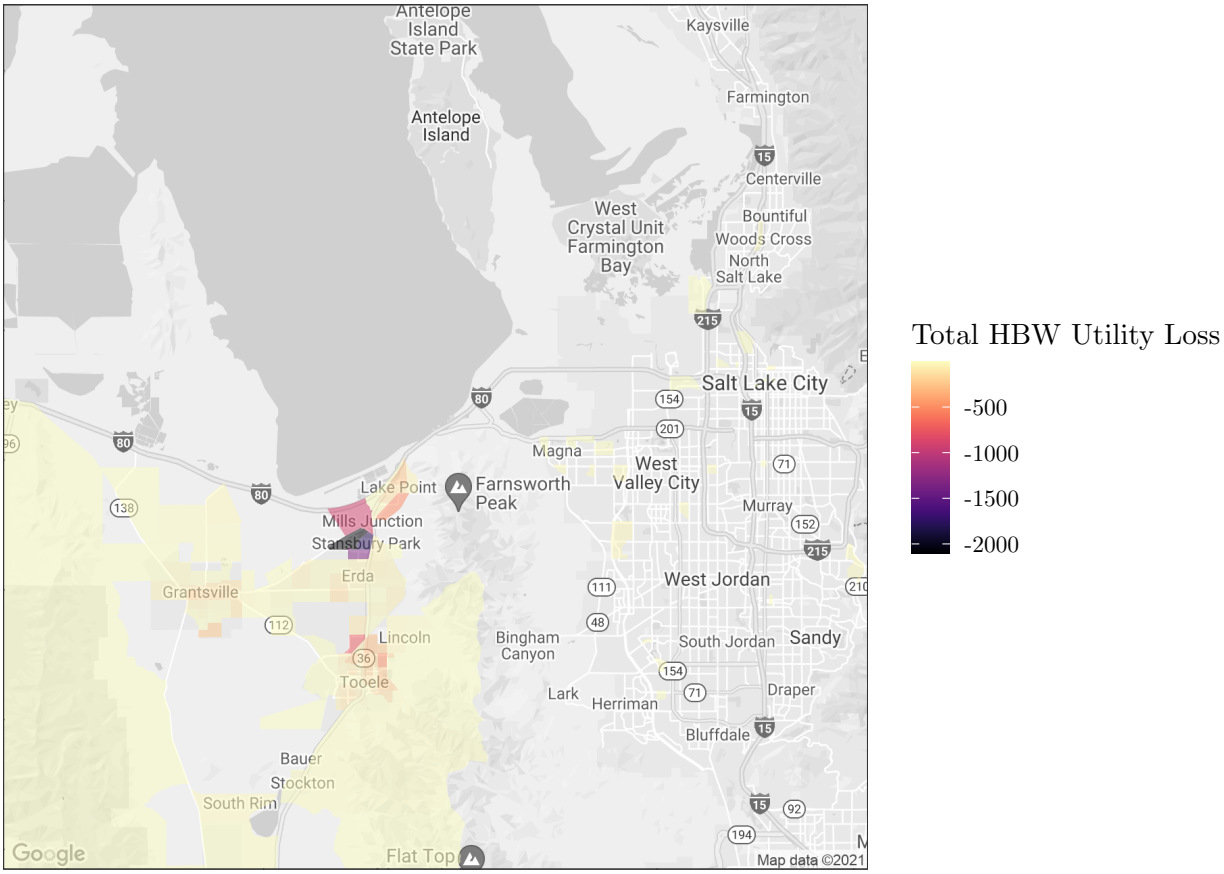

Figure 2: Total loss from

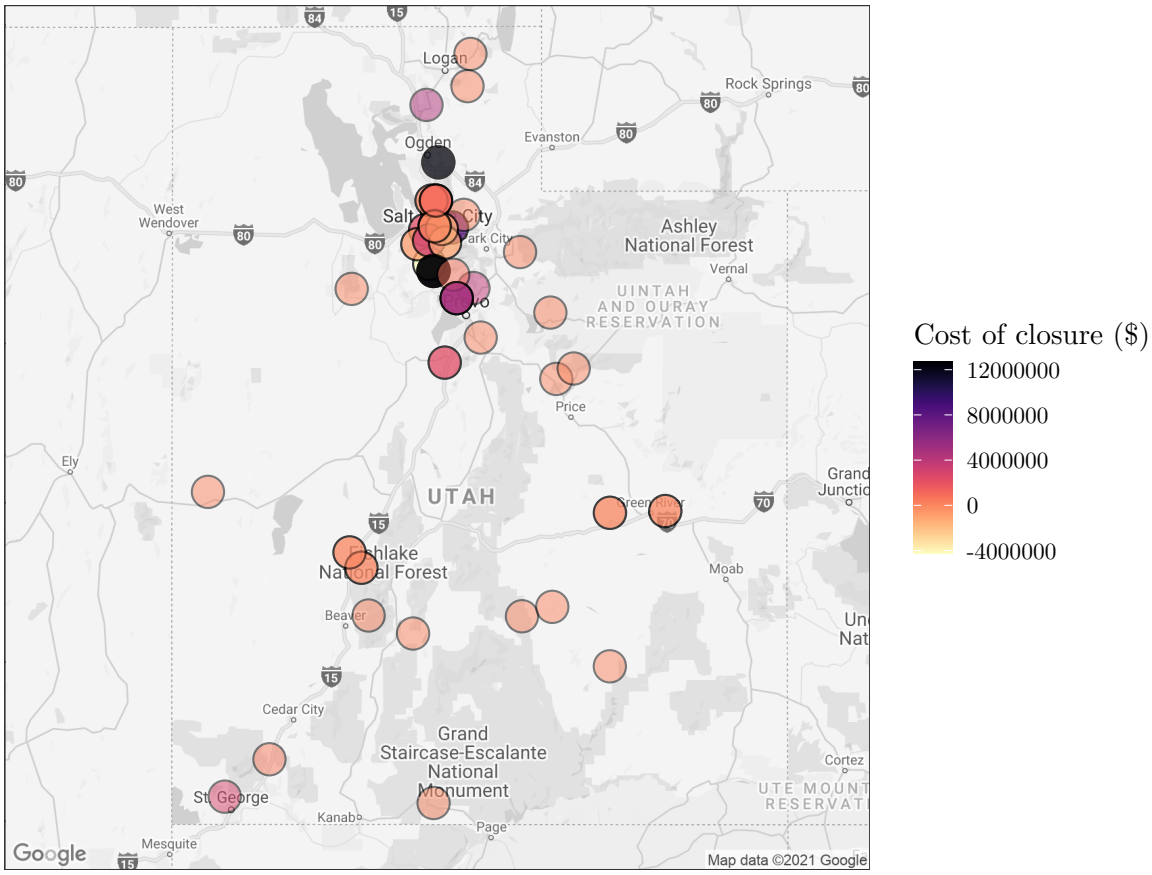

Figure 3: Total cost of link closure.

scenario, and then multiplies that difference by the VOT for each trip purpose and the number of trips estimated for each trip purpose. For external trips, freight trips, and REC trips, these were all extracted directly from USTM. The HBW, HBO, and NHB purposes are estimated using the logsum model. Calculating the costs associated with the difference in travel time allows a better estimation of the true cost experienced by road users who are not included in the logsum based resiliency model. So, while the logit-based model is more sensitive to changes in the network, it is not able to truly estimate the costs associated with closure for those purposes that are not included in the analysis.

4.3. Comparative Scenario Results

We now apply the model to compare 40 additional scenarios where individual highway facilities are removed from the model highway network. These scenario locations are shown in Figure **??**, and were identified in a report by AEM (2017). AEM approached the idea of systemic resiliency by attempting to classify various types of threats toward specific infrastructure types. This method, while valid, was difficult to implement statewide. Some facilities have obvious natural threats such as earthquake, flood, or landslide.

This section will present the results of the 41 scenarios analyzed. First, Table 5 shows each of the scenarios, labeled "road10" for scenario 10, and "road11" for scenario 11 along with route numbers or street names, as well as a physical location where the link was cut.

The logsum method results are as follows in Table 14. The results are ranked from the road with the largest cost to the road with the smallest cost. We can see that road 50, which corresponds to I-80 between SLC and Tooele, experiences the largest cost per day according to the resiliency model. Following road 50, roads 38, 27, 37, and 14 make up the five most important road according to the cost estimation provided

Table 5: Link Table

LINK_ID	ROUTE	LOCATION
road10	SR-95	near Hite
road11	US-6	near King Top
road12	I-15	in Bountiful
road13	I-70	at Dragon Point (W of Green River)
road14	I-15	in Orem between Univ. Ave & Center St
road15	SR-199	near Rush Valley
road16	SR-153	between Beaver & Junction
road17	SR-18	just North of St. George
road18	I-15	near Rocky Ridge (between Payson & Nephi)
road19	I-15	near I-70 & Filmore
road20	I-15	near New Harmony
road21	US-40	East of Strawberry Reservoir
road22	US-6	in Carbon County North of Helper
road23	Legacy Parkway	near West Bountiful
road24	UT-35	outside of Francis
road25	Timp Highway	at the base of AF Canyon
road26	SR-14	in Cedar Canyon
road27	I-84	between Ogden and Morgan
road28	SR-65	on the border of Salt Lake County & Morgan County
road29	SR-101	East of Hyrum
road30	US-91	between Brigham City & Mantua
road31	SR-62	East of Kingston
road32	US-89	between Logan and Bear Lake
road33	SR-24	in Capitol Reef National Park
road34	Bangerter	near Bluffdale
road35	SR-191	between Helper & Duchesne
road36	SR-24	near Steamboat Point
road37	I-80	in Parley's Canyon
road38	I-15	at the Point of the Mountain
road39	I-80	in SLC near Sugar House and 1300 E
road40	I-15	in SLC between 2100 S & 1300 S
road41	I-215	near Taylorsville
road42	Bangerter	near West Valley City
road43	I-215	near Cottonwood Heights
road44	MVC (UT-85)	West of West Jordan
road45	US-89	near the Border of Arizona by Lake Powell
road46	SR-189	up Provo Canyon near Vivian Park
road47	US-6	up Spanish Fork Canyon near Diamond Fork Rd
road48	I-70	near Green River (NW of Moab)
road49	I-70	near Richfield & Filmore
road50	I-80	between SLC and Tooele

by the resiliency model. Each of these roads is an Interstate facility in Northern Utah, which is heavily populated. Some roads, such as road 10, which is along SR-95 near Hite, experiences no measurable change to HBW, HBO, or NHB traffic. This is likely due to the remote location of the highway link that was cut. A ranking is provided for all of the roads in Table **??**.

Table **??** contains the ranking results from the travel time method. Here, we see that the first five roads differ from the results of the logsum model. Instead, road 48 becomes the most important road due to costs experienced. Road 48 is a part of I-70 near Green River, Utah. The other four roads that make up the top five most important roads in the travel time method analysis are road 13, road 20, road 50, and road 49. Some of these roads appear in both the logsum and travel time methods. Here again, several of the roads that are most important are located in Northern Utah. It is important to note that the main driving factor as to why a road was important or not in the travel time analysis was how much freight and external traffic it experienced. Including the freight, even with the logsum results, changes the rankings drastically.

Some other interesting findings are that in the top 10 of each analysis methods, three scenarios appear in both rankings. Road 50, which corresponds to I-80 between Tooele and SLC, road 27 which corresponds to I-84 in Weber Canyon, and road 37 which corresponds to I-80 in Parley's Canyon are included in the top 10 scenarios for both methods of analysis. This is likely due to the number of passenger trips along these routes and the number of freight trips that occur along these routes.

4.4. Positive Benefit Scenarios

Five of the scenarios indicated a benefit resulting from highway link closure. These scenarios were examined more closely to determine what causes these atypical results. The affected links are all located in Salt Lake Valley at the following locations: Bangerter Highway near Bluffdale, I-80 near 1300 E, I-15 between 2100 S and 1300 S, I-215 near Cottonwood Heights, and Mountain View Corridor near West Jordan.

It was determined that a likely cause which explains these atypical results is that the shortest path by time is not the shortest path by distance. The automobile accessibility is determined by the AM congested travel time in the Utah Statewide Travel Model (USTM). The travel distance – used to determine the accessibility of destinations by walking – is the distance of that path, and not the actual shortest distance path as might be more preferable.

When a highway link is broken, the new shortest path by time is longer than in the base scenario with this link available. But the new path may actually be shorter by distance. This causes an increase in the utility of accessing destinations by non-motorized modes, potentially overwhelming the decrease in automobile utility.

This is only observed in heavily urbanized regions for two reasons:

- The presence of high-speed expressways and parallel local roads means that alternate paths with shorter distances but longer vehicle times are more likely.
- The increased availability of destinations within the non-motorized distance threshold (50 miles) means that alternative destinations exist.

Overall, the results of the analysis indicate that the likely cause of a positive cost being estimated for these five scenarios is that there are easily accessible alternate routes in the area, along with competing TAZ of similar size in the DC size term equation.

4.5. Summary

The overall results show that the resiliency model is more sensitive to network changes than the travel time comparison. The ability for a user to choose both a mode and destination (or alternate destination) cause the logsum results to often estimate a smaller cost than the travel time results would. However, when the travel time results are factored in, the overall rankings of the 41 scenarios considered change dramatically. This is due to the large expenses experienced by freight traffic, which has a much higher VOT than other passenger trips do. In summary, Table **??** and Table **??** show the rankings for both the logsum and travel time analysis respectively. The logsum suggests that I-80 between Tooele and SLC is the most important road, while the travel time method, or total priority, indicates that I-70 near Green River is the most important road due to cost associated with closure.

Table 6: Total cost in dollars

Rank	Scenario	Delta	Cost Value	ROUTE	LOCATION
1	ROAD50	-27839.300	173995.82	I-80	between SLC and Tooele
2	ROAD38	-20277.800	126736.38	I-15	at the Point of the Mountain
3	ROAD27	-19625.700	122660.89	I-84	Weber Canyon
4	ROAD37	-13163.500	82272.039999999994	I-80	in Parley's Canyon
5	ROAD14	-7682.830	48017.71	I-15	in Orem between Univ. Ave & Center St
6	ROAD30	-7079.660	44247.87	US-91	between Brigham City & Mantua
7	ROAD46	-6009.630	37560.19	SR-189	up Provo Canyon near Vivian Park
8	ROAD41	-5154.680	32216.74	I-215	near Taylorsville
9	ROAD17	-4339.910	27124.45	SR-18	just North of St. George
10	ROAD18	-3717.130	23232.06	I-15	near Rocky Ridge (between Payson & Nep
11	ROAD42	-3665.700	22910.63	Bangerter	near West Valley City
12	ROAD12	-1588.610	9928.7900000000009	I-15	in Bountiful
13	ROAD25	-1092.820	6830.11	Timp Highway	at the base of AF Canyon
14	ROAD20	-476.542	2978.39	I-15	near New Harmony
15	ROAD24	-257.283	1608.02	UT-35	outside of Francis
16	ROAD47	-220.368	1377.3	US-6	up Spanish Fork Canyon at Diamond Forl
17	ROAD23	-159.863	999.14	Legacy Parkway	near West Bountiful
18	ROAD26	-119.912	749.45	SR-14	in Cedar Canyon
19	ROAD15	-80.225	501.41	SR-199	near Rush Valley
20	ROAD32	-66.768	417.3	US-89	between Logan and Bear Lake
21	ROAD31	-39.820	248.88	SR-62	East of Kingston
22	ROAD49	-27.569	172.31	I-70	near Richfield & Fillmore
23	ROAD21	-26.323	164.52	US-40	East of Strawberry Reservoir
24	ROAD29	-25.200	157.5	SR-101	East of Hyrum
25	ROAD19	-22.071	137.94	I-15	near I-70 & Filmore
26	ROAD22	-21.230	132.69	US-6	in Carbon County North of Helper
27	ROAD45	-10.441	65.260000000000005	US-89	near the Border of Arizona by Lake Powel
28	ROAD16	-7.775	48.59	SR-153	between Beaver & Junction
29	ROAD48	-5.883	36.770000000000003	I-70	near Green River (NW of Moab)
30	ROAD35	-5.713	35.71	SR-191	between Helper & Dechesne
31	ROAD36	-5.126	32.04	SR-24	near Steamboat Point
32	ROAD28	-5.049	31.56	SR-65	border of Salt Lake & Morgan Counties
33	ROAD33	-0.201	1.26	SR-24	in Capitol Reef National Park
34	ROAD13	-0.012	7.0000000000000007E-2	I-70	at Dragon Point (W of Green River)
35	ROAD10	0.000	$ -	SR-95	near Hite
36	ROAD11	0.000	$ -	US-6	near King Top
37	ROAD40	29.891	-186.82	I-15	in SLC between 2100 S & 1300 S
38	ROAD44	1191.213	-7445.08	MVC (UT-85)	West of West Jordan
39	ROAD43	1812.615	-11328.84	I-215	near Cottonwood Heights
40	ROAD39	2541.037	-15881.48	I-80	in SLC near Sugar House and 1300 E
41	ROAD34	6822.414	-42640.09	Bangerter	near Bluffdale

Table 7: Total cost in dollars

Rank	Scenario	HBO	HBW	NHB	Freight	XXP	TOTAL
1	ROAD48	2.31	18.06	16.40	85537717.30	417991.55	85955745.62
2	ROAD13	0.00	0.00	0.07	25214668.62	95273.29	25309941.98
3	ROAD20	873.21	1008.53	1096.65	19285468.81	234813.54	19523260.75
4	ROAD50	27301.63	106106.36	40587.84	10883835.01	55870.17	11113701.00
5	ROAD49	15.89	28.48	127.94	8118515.31	29826.77	8148514.39
6	ROAD18	5278.85	8534.44	9418.77	5784337.92	143890.39	5951460.37
7	ROAD27	56096.79	39697.31	26866.79	4545561.25	9608.65	4677830.80
8	ROAD19	24.73	17.79	95.43	3377211.54	94820.44	3472169.92
9	ROAD37	22074.06	34804.83	25393.16	2060291.60	5668.64	2148232.29
10	ROAD47	89.03	868.43	419.84	2028078.38	14011.56	2043467.24
11	ROAD38	21901.51	50046.84	54788.03	1732758.70	17292.88	1876787.96
12	ROAD14	-259.88	25793.30	22484.29	950455.53	10899.55	1009372.79
13	ROAD30	17333.42	13857.91	13056.54	911254.89	3690.26	959193.02
14	ROAD39	-19335.84	5106.11	-1651.75	830874.97	2821.03	817814.52
15	ROAD22	44.08	11.04	77.57	682605.02	4843.35	687581.05
16	ROAD45	14.58	24.91	25.77	551950.51	40330.43	592346.20
17	ROAD21	34.71	16.42	113.39	537843.44	3397.73	541405.69
18	ROAD12	453.26	3435.59	6039.94	377513.25	3445.92	390887.95
19	ROAD40	-5542.43	6773.59	-1417.99	370214.60	3233.35	373261.13
20	ROAD35	24.84	0.00	10.87	181286.62	2691.10	184013.42
21	ROAD46	10491.62	12149.38	14919.20	81734.81	171.87	119466.87
22	ROAD32	55.04	206.03	156.24	48524.89	0.00	48942.19
23	ROAD41	4930.44	11871.22	15415.08	15154.31	0.00	47371.04
24	ROAD16	13.22	26.42	8.96	30533.27	0.00	30581.87
25	ROAD17	15003.63	4553.69	7567.12	2085.51	0.00	29209.96
26	ROAD42	4150.58	11518.83	7241.23	4320.63	0.00	27231.26
27	ROAD11	0.00	0.00	0.00	7576.95	0.00	7576.95
28	ROAD25	5099.17	1177.88	553.06	56.27	0.00	6886.38
29	ROAD33	0.00	1.15	0.11	4089.04	0.00	4090.30
30	ROAD24	824.28	498.41	285.33	1884.43	0.00	3492.45
31	ROAD36	9.06	4.02	18.96	1920.11	0.00	1952.15
32	ROAD26	227.74	302.12	219.59	733.26	0.00	1482.71
33	ROAD23	574.39	154.86	269.89	168.23	0.00	1167.37
34	ROAD15	47.73	208.09	245.59	209.74	0.00	711.15
35	ROAD31	32.86	95.94	120.08	259.51	0.00	508.38
36	ROAD10	0.00	0.00	0.00	345.94	0.00	345.94
37	ROAD29	37.13	119.70	0.67	1.93	0.00	159.43
38	ROAD28	0.00	0.21	31.34	5.78	0.00	37.34
39	ROAD44	-8946.88	2539.77	-1037.98	3495.97	0.00	-3949.11
40	ROAD43	-11917.37	2436.49	-1847.96	4930.45	0.00	-6398.40
41	ROAD34	-21929.15	-4681.46	-16029.48	19511.32	0.00	-23128.77

5. Conclusions

The USTM model is a gravity-based travel demand model, while the resiliency model is logit-based. The logit-based nature of the resiliency model allows for greater sensitivity in user mode and destination choice, which causes the estimated costs associated with link closure to be lower than for previous estimations. The resiliency models incorporation of both the logsum for HBW, HBW, and NHB purposes, as well as the travel time calculation for the other purposes included in USTM. The recommendations resulting from the adaptation of a logit-based model on the USTM network are that logit-based modeling returns more sensitive estimations of the value of a link in the network. This is a highly important outcome because more accurate estimation allows UDOT to better understand the monetary importance of highway links throughout Utah. Additionally, the model's design allows for further analysis of additional link closures, or even multi-link closure.

5.1. Summary

The development of a logit-based travel demand model can improve the ability of UDOT to accurately estimate the costs per day of link loss that Utahn's would experience. The resiliency model provides sensitive estimates that more accurately represent the costs associated with link closure than a travel time increase methodology by itself would be able to capture. Using the logsum, estimations sensitive to user choice were found, which can help professionals to better evaluate risk to Utah's infrastructure. User choice is highly important in modern modeling practices because user choice allows a model to estimate information more precisely and accurately. The information provided by the resiliency model should be used to prioritize link importance to the functionality of Utah's highway network. The resiliency model more accurately estimates costs experienced by Utahn's due to link loss than do traditional methods for determining costs.

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